

Paper Practice 3.2**Functions**

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. For every yes-or-no question, explain your reasoning in a complete sentence.

- 1.** Determine whether each relation is a function.

Explain. (Relations)

a. $\{(2, 5), (4, 7), (8, 11), (4, 13)\}$

ANSWER: No – the input 4 is paired with both 7 and 13.

b. $\{(-3, 6), (0, 6), (5, 6), (9, 6)\}$ ANSWER:
Yes – every input appears once. A repeated output is allowed.

c. $\{(1, 2), (2, 4), (3, 6), (2, 8)\}$ ANSWER:
No – the input 2 is paired with both 4 and 8.

- 2.** State the domain and the range of each relation. (Domain & Range)

a. $\{(-4, 3), (-1, 0), (2, 5), (6, 3)\}$

b. $\{(0, -2), (1, -2), (2, -2)\}$

- 3.** A mapping pairs $3 \rightarrow 7$, $5 \rightarrow 7$, $8 \rightarrow 9$, and $3 \rightarrow 11$. Is the relation a function? Explain. (Mapping)
ANSWER: No – 3 is paired with both 7 and 11, so one input has two outputs.

- 4.** If $f(x) = 3x - 7$, find each value. (Function Notation)

a. $f(4)$

b. $f(-2)$

c. $f(0)$

- 5.** If $g(x) = x^2 + 2x$, find each value. (Function Notation)

a. $g(3)$ ANSWER: $g(3) = 9 + 6 = 15$

b. $g(-5)$ ANSWER: $g(-5) = 25 - 10 = 15$

c. $g(0)$ ANSWER: $g(0) = 0$

- 6.** If $f(x) = 2x + 9$, find the value of x for which $f(x) = 21$. (Solve for the Input)

- 7.** A vertical line drawn at $x = 2$ crosses a graph at $(2, 1)$ and at $(2, -3)$. Is the graph a function? Explain. (Vertical Line Test) ANSWER: No – it fails the vertical line test. The input 2 has two outputs.
- 8.** Madison took five quizzes. Her scores were Quiz 1: 86, Quiz 2: 92, Quiz 3: 78, Quiz 4: 86, Quiz 5: 84. (Analyze Data)
- What is the domain?
 - What is the range?
 - Is this relation a function? Explain.
- 9.** The table below shows a relation. Is it a function? Explain. (Table)
- x : $-2, -1, 0, 1$ y : $5, 5, 5, 5$
ANSWER: Yes – every input appears once. Outputs may repeat.
- 10.** Error Analysis – Gus says $\{(1, 4), (2, 4), (3, 4)\}$ is not a function because the value 4 repeats. Identify his mistake, then state whether the relation is a function. (Error Analysis)
- 11.** Write the relation $\{(1, 3), (2, 6), (3, 9), (4, 12)\}$ as a rule in function notation. (Write a Rule)
ANSWER: $f(x) = 3x$
- 12.** Write a set of four ordered pairs that is NOT a function, then explain in one sentence what makes it fail. (Construct)